

A Project Report on

# **RFID Based Door Lock Security System**

A Real Time Project Report submitted in a partial fulfilment of the requirement  
for the award of the degree of

**BACHELOR OF TECHNOLOGY**  
**(Integrated Dual Degree Program)**

In

**ELECTRONICS AND COMMUNICATION ENGINEERING**

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**DECLARATION OF THE CANDIDATES**

We, the undersigned—K.B. S Karthikeya (22011P0410), N. Saikiran (22011P0412), G. Abhiram (22011P0427), M. Sadvika (22011P0442), and A. Aishwarya (22011P0449)—hereby affirm that the project report titled “**RFID Based Door Lock Security System**” represents an authentic record of our work. This report was completed under the esteemed guidance of Mrs. **Samrajam Jyothula**, Assistant Professor in the Department of Electronics and Communication Engineering at JNTUH.

The project fulfils the requirements for the award of the **Bachelor of Technology (Integrated Dual Degree Program)** degree in **Electronics and Communication Engineering** during the academic year 2023-2024. The results presented herein are original and have not been copied from any other source, nor have they been submitted to any other university or institute for the award of any other degree or diploma.

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**CERTIFICATE BY THE SUPERVISOR**

This is to certify that this is a Bonafide record of the dissertation work entitled, “RFID Based Door Lock Security System” , done by —K.B.S Karthikeya (22011P0410), N. Saikiran (22011P0412), G. Abhiram (22011P0427), M. Sadvika (22011P0442), and A. Aishwarya (22011P0449)—in partial fulfilment of the requirements for the award of the degree of **Bachelor of Technology (Integrated Dual Degree Program)** in Electronics and Communication Engineering from JNTUH during the academic year 2023-2024, is a genuine record of their work conducted under the supervision of **Mrs. Samrajam Jyothula, Assistant Professor**. This dissertation has not been submitted to any other university or institution for the award of any other degree. The results have been thoroughly verified and found to be satisfactory.

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## **ABSTRACT**

In recent times even though, cybercrimes have increased we can't say offline theft is gone completely. A well secured house, offices, lockers are important for the safety of property. Even though there are locks to doors they do not provide much security since keys can be duplicated easily. Nowadays a lot of new technologies have emerged to overcome the drawbacks of traditional door locking systems. These alternatives not only help to keep the house secure but also allows for remote access of the door with just one click. One of those is Radio frequency identification (RFID).

A RFID door lock prevents the access of unauthorised persons. When the RFID system combined with Internet of Things (IOT) technologies provide even more security. Our door lock system makes use of Global System of Mobile Communication to provide a message or mail alert to the owner. This technology is useful in times of intrusion, the message alert can help assess the situation and respond accordingly.

To eradicate the usage of computationally expensive systems, the proposed system uses the Arduino uno, which is simpler to deploy, easier to integrate and require minimal time to perform the desired operation. This can be implemented in educational institutes, industrial organizations, hospitals, etc. to provide security. This wireless system provides security way more efficiently than the normal door locks which makes it a great choice. This system is a scalable, sizable, cost-efficient and efficient way to have a door lock system.

# MOTIVATION

The door security has always been an integral part of human life doors aren't necessarily only for homes they may extend their use to lockers, cars, community halls and many more. There are some places which hold important things that need to be protected this is where RFID based door lock system provides extra security, and our motivation is from here.

People have already realised the importance of the door lock and provided us with many different door lock systems on complex than another some are just manual locks which need pattern recognition or have tricks. Slowly these door lock systems turned from mechanical to electric and started using keypads and tags and different mechanisms.

These mechanisms led into integration of each other to build complex door locks.

One of such is the RFID and keypad-based system.

The system used a RFID detection and password system which is a soundly secure system. We had the intent to upgrade the system to another level, so we thought of the recent trend the OTP (One Time Password) and there sparked the idea of using an OTP instead of the password to make the game of door locks more secure.

This represents an opportunity for improvement and as engineers we must utilise the scope and provide a stable system which is useful to society and science.

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# INTRODUCTION

## Objective

The aim of this project is to develop a highly secure door lock and alert system utilizing both Radio Frequency Identification (RFID) and a keypad which has and integrating it with an Arduino microcontroller.

## Brief

The objective of an RFID and keypad-based door lock and alert system using Arduino is to create a secure and user-friendly access control system for doors. This system would integrate RFID technology and a numeric keypad to manage and control entry, along with an alert mechanism to enhance security.

## Literature Review

Access control systems have witnessed a remarkable evolution from their inception, shifting from traditional mechanical locks to sophisticated electronic solutions that enhance both security and user convenience. Among these advancements, RFID (Radio-Frequency Identification) systems combined with static passwords and those integrated with OTP (One-Time Password) technology stand out as significant innovations in modern security management.

### **Traditional RFID-Based Systems with Static Passwords**

RFID technology has revolutionized access control by enabling contactless authentication. In traditional systems, users are granted access by presenting an RFID tag to a reader, which verifies their identity against a pre-established database. To add an extra layer of security, these systems often incorporate a static password or PIN code that users must enter via a keypad.

The primary advantage of RFID-based systems with static passwords lies in their convenience. The contactless nature of RFID technology simplifies access, eliminating the need for physical keys and reducing the time required for authentication. Users only need to present their RFID tag and input a static password to gain entry, which is straightforward and easy to manage.

However, this approach is not without its drawbacks. The reliance on static passwords introduces significant security vulnerabilities. Static passwords are susceptible to various types of attacks, including brute force attacks, credential

theft, and unauthorized sharing. If an attacker gains access to the RFID tag or the password, they can potentially compromise the entire security system. Additionally, managing and updating static passwords can be cumbersome, particularly in large systems with numerous users. This complexity can lead to potential security lapses and administrative challenges.

### **Advancing with RFID and OTP Integration**

In contrast, the advanced RFID system integrated with OTP technology represents a significant leap forward in access control security. In this system, the process begins with the user presenting their RFID tag to the reader, which authenticates their identity. Following this, an OTP is generated and sent to the user's mobile device. To gain final access, the user must enter this OTP into the system.

The integration of OTP technology addresses many of the security concerns associated with static passwords. OTPs are dynamic and temporary, meaning they are valid for only a short period and are unique for each authentication attempt. This makes them highly resistant to interception and reuse. Even if an attacker compromises the RFID tag or the OTP, the risk of unauthorized access is minimized due to the temporary nature of the OTP. This advanced security measure significantly enhances the overall protection provided by the system.

# **LIST OF COMPONENTS**

To solve the above problem, we introduce our solution RFID based door lock with GSM integration to make our lives much safer and reduce our concerns about security.

## **Software requirements**

Arduino Integrated Development Environment (IDE)

## **Hardware requirements**

1. Arduino UNO
2. I2C LCD
3. MFRC522 RFID Reader and Tags
4. SIM900 GSM module
5. 5V, 2A power adapter
6. SG90 Micro-servo motor
7. 4X4 Keypad
8. Resistor 220 ohm
9. Buzzer
10. 6 to 12V Power source
12. Breadboard
13. Jumper wires
14. LED



# **Radio-frequency identification (RFID)**

## **Introduction**

RFID is a wireless communication technology that employs electromagnetic waves to identify and track tags attached to objects, people, or animals. These tags store digitally encoded data, which can be read by an RFID reader. Unlike barcode readers, RFID readers do not require direct line-of-sight with the tag, allowing them to read tags from a distance.

In 1945, Leon Theremin, a Soviet inventor, created a remarkable device known as the “Thing.” the Thing played a pivotal role in the evolution of RFID technology. Here’s a concise overview:

The Thing: Leon Theremin’s invention was a covert listening device. It retransmitted incident radio waves, adding audio information. When sound waves vibrated a diaphragm, they subtly altered the resonator’s shape, which in turn modulated the reflected radio frequency. Importantly, the Thing was passive—it drew energy from external waves to activate itself.

While the Thing served a different purpose, its passive nature and interaction with radio waves foreshadowed the principles underlying modern RFID systems. These systems have since transformed various industries, from inventory management to retail and beyond [28].

## **RFID PRINCIPLE [2]**

Source: S. Guizani, "Security applications challenges of RFID technology and possible countermeasures," 2014 International Conference on Computing, Management and Telecommunications (ComManTel), Da Nang, Vietnam, 2014, pp. 291-297, doi: 10.1109/ComManTel.2014.6825620. keywords: {Radiofrequency identification;Antennas;ISO standards;Security;Communication channels;Integrated circuits;RFID;Product Code;Management;Intelligent systems},

RFID technology hangs on the idea of assigning a unique number to each individual item (object, animal, human being). To setup an RFID system, three main components are required

The tag: also called transponder, is an integrated circuit, commonly referred to as microchip, attached to a minuscule (metallic) antenna. Every tag has its own identifier code that will be used to uniquely identify the item associated to it.

The reader: is used to create an interrogation zone in which the RFID tag will be read. It collects information from the tags, filter them and transfer them to the processing unit. In the case of a passive tag, the reader also serves to power the tag up. Moreover, the reader manages the different antennas it is connected to achieve a maximum read rate.

The server / host is the processing part of the RFID system. The host is a computer that runs, in addition to the enterprise application, an application called middleware that is an interface between the reader and the application layer. The middleware, the brain behind the smart system, manages the different readers and instructs each one on how to interact with the RFID transponders. It is also the component that is responsible for communicating filtered events back to the application. The host may be one processing unit or an entire enterprise system composed of servers (web, databases...). The host may be one processing unit or an entire enterprise system composed of servers (web, databases...)

The operating principle of an RFID system is the following. The reader starts a communication process by emitting electromagnetic wave. This wave will be intercepted by the antenna of the RFID tag, generally a passive tag. An induced current will activate the integrated circuit, enabling it to communicate. To transmit data stored in its memory (chip), the tag modulates the digital signal using phase or amplitude modulation on a predefined frequency carrier. The reader receives information and redirects it to the host where it will be processed.

## **RFID SYSTEM CLASSIFICATION**

The RFID systems can be classified based on two criteria namely:

the operating frequency of the system

the power source of the tags

### **Frequency allocation of the RFID**

Frequency is a major factor in RFID. The characteristics of the electromagnetic wave change from a band to another.

We also note that the HF domain remains the most used with 1605 (=1320+285) case studies in 2007 out of 2066 projects. That is almost 78% of market size as

presented by the following pie chart. By the end of June 2010, the case study numbers have reached 4050 with 4100 company covered in 111 countries.

An RF signal can be efficiently radiated if the size of the antenna is comparable to the wavelength of the operating frequency.

| TYPES        | LOW FREQUENCY                                                                                                                                            | HIGH FREQUENCY                                                                                                                                                                       | ULTRA HIGH FREQUENCY                                                                                                                                                          |
|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| RANGE        | Few centimetres                                                                                                                                          | 50cm                                                                                                                                                                                 | 6m                                                                                                                                                                            |
| ADVANTAGES   | <ul style="list-style-type: none"> <li>- non affected by water</li> <li>- non affected by metals</li> <li>- frequency use without restriction</li> </ul> | <ul style="list-style-type: none"> <li>- non affected by water</li> <li>- non affected by metals</li> <li>- multiple tag read</li> <li>- non affected by electrical noise</li> </ul> | <ul style="list-style-type: none"> <li>- long range</li> <li>- standard</li> <li>- high rate</li> <li>- easy to produce with low cost (5 cents)</li> </ul>                    |
| DRAWBACKS    | <ul style="list-style-type: none"> <li>- expansive</li> <li>- noise</li> <li>- low rate (70 ms to read one tag)</li> </ul>                               | <ul style="list-style-type: none"> <li>- range &lt; 1 m</li> <li>- less efficient than LF (water and metals)</li> </ul>                                                              | <ul style="list-style-type: none"> <li>- absorbed by water</li> <li>- reflected by metals</li> <li>- limited memory</li> <li>- interference with many applications</li> </ul> |
| APPLICATIONS | <ul style="list-style-type: none"> <li>- animal tracking</li> <li>- identification</li> </ul>                                                            | <ul style="list-style-type: none"> <li>- credit card</li> <li>- access control card</li> <li>- passports</li> </ul>                                                                  | <ul style="list-style-type: none"> <li>- industry</li> <li>- retail chain</li> </ul>                                                                                          |

Table 1

S. Guizani, "Security applications challenges of RFID technology and possible countermeasures," 2014 International Conference on Computing, Management and Telecommunications (ComManTel), Da Nang, Vietnam, 2014, pp. 291-297, doi: 10.1109/ComManTel.2014.6825620. keywords: {Radiofrequency

identification;Antennas;ISO standards;Security;Communication channels;Integrated circuits;RFID;Product Code;Management;Intelligent systems}.

However, for 13.56 MHz frequency the expected antenna size is about 22.12 meters, which is not feasible. As alternative, a small loop antenna circuit, resonating at the operating frequency, is used.

Compared to HF (13.56MHz) and Microwave (2.45GHz), the basic advantage of UHF passive RFID system is its long-range communication between the reader and tags.

### **Tag power source classification**

An RFID tag is composed of a simple silicon microchip combined with a metallic antenna in a compact package. It can be as small as a grain of sand or as big as a book. There are three main families used to classify tags according to their energy source and functionalities: active tags, passive tags, or semi-passive tags

Active tags have a battery included in the transponder and actively transmit on longer distances reaching several kilometres. These tags are in general larger, more expensive and more durable. They are mainly used for tracking trailers in yards and containers on loading docks.

Passive tags extract their power from the electromagnetic waves emitted by the reader's antenna. Typically, passive RFID tags are uses as ID badge at work, automatic access card, speed pass to purchase gasoline or as protection in cars having an antitheft immobilizer.

Semi-passive tags are like passive ones in the sense that they extract their power from the reader for communication purposes. However, they also have a built-in battery that allows them to store data on the microchip. The battery is not used for powering the tag. Generally, these tags include sensors, which need batteries.

## **SECURITY ASPECTS AND ITS COUNTERMEASURES**

Security is a crucial issue that must be addressed accordingly and since the technology is progressing daily, hacking techniques also are progressing and sometimes on a faster scale. For example, in 2008 researchers from Radboud

University in Nijmegen have developed a relatively easy method to hack and reproduce a large number of smart cards. Fake Biometric passports are also doable.

In an RFID system, reproduction of tags is quite easy. Culprits need to read the tag and to write the identifier on a different tag. Affixing a password could improve security but it does not make the system fully secure since techniques of researching passwords are very developed.

As stated previously, and like any other information systems, RFID are susceptible to outside attacks and can be altered at different stages during their employment. All attacks waged against any RFID system can be classified generally into three main groups (Attacks on CIA):

Attacks on confidentiality,

Attacks on integrity,

Attacks on authenticity

As well being vulnerable to ordinary attacks such as denial of service, eavesdropping and man-in-the-middle. Figure 5 shows that RFID technology is susceptible to spoofing and power attacks. Some kinds of different attacks are explained below, and the necessary countermeasures are provided as a way of protection and safeguard.

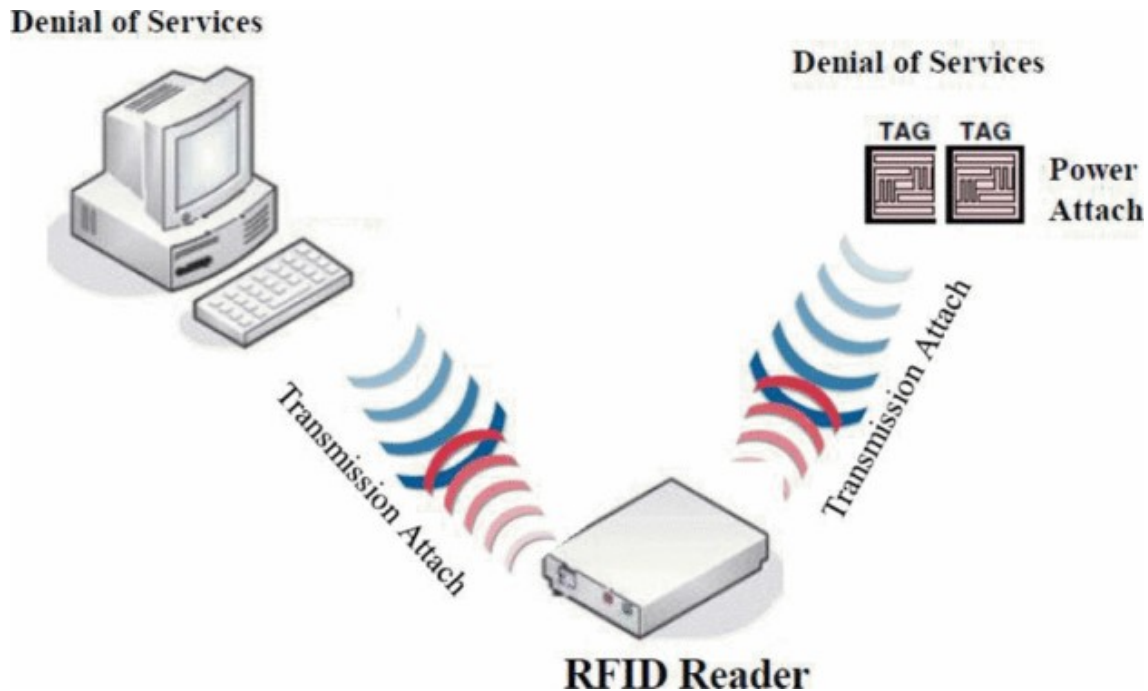


Figure 1 Denial of service attack

S. Guizani, "Security applications challenges of RFID technology and possible countermeasures," 2014 International Conference on Computing, Management and Telecommunications (ComManTel), Da Nang, Vietnam, 2014, pp. 291-297, doi: 10.1109/ComManTel.2014.6825620. keywords: {Radiofrequency identification;Antennas;ISO standards;Security;Communication channels;Integrated circuits;RFID;Product Code;Management;Intelligent systems},

## Reverse engineering

Reverse engineering is the process of discovering the technological principles of a device, object, or a system through the analysis of its structure, functions and operation. Taking into consideration the privacy issues related to the biometric e-passport, it could be feasible for an attacker to obtain access of the chip and examine its memory contents optically to recover the PIN, biometric data or any personal information, etc. The technical capability and equipment required to reverse engineer such an integrated circuit can be rated at different stages from a knowledgeable individual who will be using low cost and easily available tools to a highly skilled team, using equipment not commonly available in the commercial market (Actel, 2002).

### ***Countermeasures***

As proposed by the Federal Information Processing Standard (FIPS), coatings a chip as an anti-reverse engineering method to prevent attacks is taken into consideration. Different tamper proof procedures have been developed to protect against reverse engineering attacks. For example, by adding a tamper-release layer to RFID tags, user can be alerted if a tag has been tinkered with or not.

### **Power analysis of RFID tags**

This type of attack is unique in that it requires no physical contact with the device under attack. Power analysis is a type of side-channel attack that is meant to salvage data and information by analysing changes in the power use of such device. It has been shown that the power emission patterns are different when the card received correct and incorrect password bits or cryptographic keys. It is possible to breach smart card security by monitoring power consumption signals. It is also predicted that a power analysis attack on a RFID tag could be carried out using a simple common device such as a cell phone.

### ***Countermeasures***

The widespread methods used to overcome power analysis attacks are filtering or adding an element of arbitrariness. Filtering power signals or delaying the calculation randomly can increase the difficulty for the attacker to recognize the power consumption patterns.

### **Eavesdropping**

Eavesdropping is the act of secretly listening to the private conversation of others without their consent. Given that an RFID tag is a wireless device that emits and transmits data, there always exists a risk that the communication between tag and reader can be eavesdropped when interrogated by an RFID reader. Eavesdropping will happen when an attacker captures data with an analogous reader (one with most likely with same tag type and frequency), while a tag is being read by an authorized RFID reader. As most RFID systems use clear text way of communication, due to tag low memory capacity and or cost, eavesdropping is an effortless action, but it is an efficient way for the attacker to obtain and gather useful information on the collected tag data. The information obtained during the attack can have serious implications and severe consequences, it could be used in future attacks against the same RFID system.

### ***Countermeasures***

The best action against eavesdropping does include establishing a secure channel and/or encrypting the communication between the tag and the reader

### **Man-in-the-middle attack (Mim)**

The man-in-the-middle attack (Mim) is a form of active eavesdropping in which the attacker makes independent connections with the victims and relays messages between them, making them believe that they are talking directly to each other over a private connection, when in fact the entire conversation is controlled by the attacker. The attacker can interrupt the communication pathway and manipulate the information back and forth between the RFID components. The attacker exposes the data prior to the anticipated device obtain it and can modify the information an route.

### ***Countermeasures***

There exist some techniques that can be applied to diminish Mim threats, such as encrypting data communications, transfer information through a protected channel, and afford an authentication protocol.

### **Denial of Service (DoS)**

A denial-of-service (DoS) attack is an attempt to make a machine or network resource unavailable to its intended users. Although the means to carry out, motives for, and targets of a DoS attack may vary, it generally consists of efforts to temporarily or indefinitely interrupt or suspend services of a host connected to the Internet. DoS attacks can take many forms to attack an RFID tag. The main reason is not to take or modify any data or information, but to immobilize the RFID system so that it will be disabled.

### ***Countermeasures***

Detecting DoS attacks is easier than preventing them from happening in general. Nevertheless, once the attack is detected, it normally can be stopped before it can do too much damage by implementing filters to block unwanted traffic.

### **Spoofing**

A spoofing attack is a situation in which one person or program successfully masquerades as another by falsifying data and thereby gaining an illegitimate advantage. With respect to RFID system, spoofing take place when a fictitious



tag pretence as a legitimate tag and thereby gains an illegitimate and illegal advantage. Tag duplication is a spoofing attack where the attacker captures the information from a valid tag and creates a copy of the captured sample on a blank tag.

### ***Countermeasures***

An ordinary technique to overcome a spoofing attack is to employ an RFID authentication protocol and data encryption technique, which will enhance the rate and technology needed for a victorious attack.

# Arduino (Hardware)

## Introduction

The Arduino Uno is an open-source microcontroller board developed by Arduino.cc. It's based on the Microchip ATmega328P microcontroller and was initially released in 2010. The Uno operates at 5V, has 14 digital I/O pins, 6 analog pins, and supports communication via UART, SPI, and I2C. It's a great choice for beginners in electronics and coding due to its affordability and supportive community.

## Types of Arduino boards

### Arduino UNO R3:

The Arduino UNO R3 is one of the most popular and widely used Arduino boards. It features an ATmega328P microcontroller and offers 14 digital input/output (I/O) pins. These pins can be used for connecting sensors, actuators, and other components. Additionally, it has 6 analog input pins and supports communication protocols like SPI, I2C, and UART. The UNO is beginner-friendly, affordable, and great for prototyping projects.

### Arduino Nano:

The Arduino Nano is a compact version of the UNO. It shares similar features but comes in a smaller form factor, making it ideal for space-constrained projects. The Nano can be directly plugged into a breadboard, making it convenient for rapid prototyping. It's a favourite among hobbyists and makers due to its versatility and ease of use.

### Arduino Due:

The Arduino Due stands out with its powerful ARM Cortex-M3 SAM3X8E microcontroller. It provides 54 digital I/O pins, 12 analog inputs, and a higher clock speed compared to the UNO. The Due is suitable for more complex projects that require additional processing power. It's commonly used in robotics, automation, and advanced applications.

## **Arduino Mega:**

The Arduino Mega is designed for large-scale projects. It features an ATmega2560 microcontroller, offering a whopping 54 digital I/O pins and 16 analog inputs. With its extensive memory and multiple communication interfaces, the Mega can handle multiple sensors, actuators, and displays simultaneously. It's a go-to choice for ambitious projects and applications.

## **ARDUINO UNO**

The Arduino uno is the kind of board that is best to get started in electronics and coding. Arduino uno is easy to use and program, making it one of the best boards for beginners. With technological advancement, Arduino boards are essential in developing intelligent solutions for the future. Let's now dive into the architecture of Arduino uno and understand every part of it in detail.

The Arduino uno is an open-source electronics platform comprising hardware (microcontroller boards) and software (Arduino IDE). It is designed for easy prototyping, making it popular among hobbyists, educators, and professionals. Arduino boards are equipped with various I/O pins like digital and analog pins. The Arduino IDE is the programming environment based on C/C++. Arduino finds applications in robotics, home automation, IoT, education, and scientific research. It connects to a computer via USB for programming and power, and is compatible with a variety of shields and sensors. The Arduino uno uses the Arduino IDE for programming, which supports a wide range of libraries and examples to help with development. It has the same microcontroller as the Arduino nano board. The hardware reference design is distributed under a creative common's attributions share Alike 2.5 license and is available on the Arduino website. Layout and production files for some versions of the hardware are also available. The word uno means one in Italian and was chosen to mark a major redesign of the Arduino hardware and software. Arduino uno is a low cost, flexible, and easy to use programmable open-source microcontroller board that can be integrated with other Arduino boards, Raspberry Pi boards and can control relays, LEDs, servos, and motors as an output.

Arduino is widely used in IoT applications for collecting sensor data, processing it, and transmitting it to the cloud for analysis and control. The Arduino

programming language is used to program microcontroller boards such as the Arduino uno to interact with sensors.

## **Hardware Structure of Arduino Uno**

Arduino Uno board is an easy-to-use board that is beginner friendly. It consists of a microcontroller chip which is made up of an ATmega328p processor. It has input and output pins, Analog pins, a USB connection, a power jack and a reset button. The main component of the Arduino Uno board includes the following:

### **Microcontroller**

Arduino Uno Board is an open-source microcontroller board powered by an ATmega328p processor.

### **Input/Output Pins**

It comprises 14 input and output pins with 6 analog input pins. These pins are used to connect Arduino boards to different electronic components.

### **Power Supply**

The power supply components include IOREF, GND, 3.3V, 5V, and Vin, which connect sensors. Voltage regulators regulate the Power supply of the Arduino Uno board.

### **LED**

LED indicators are used in Arduino Uno, which is used for debugging or indicating a signal.

### **Reset Button**

A reset button is used to restart the microcontroller.

### **USB Interface:**

It consists of a USB port to upload the program to the Arduino uno board.

### **Voltage Regulator:**

It provides a stable and reliable voltage supply to the board's components.

### **ICSP Pin**

It programs the microcontroller and performs functions like debugging the code.

## **Crystal Oscillator**

It is an electronic oscillator that is used to create stable and precise frequency signals.

# Arduino Integrated Development Environment

## (Software)

**Source: Using the Arduino Software (IDE) | Arduino Documentation**

The Arduino Software (IDE) makes it easy to write code and upload it to the board offline. We recommend it for users with poor or no internet connection. This software can be used with any Arduino board.

There are currently two versions of the Arduino IDE, one is the IDE 1.x.x and the other is IDE 2.x. The IDE 2.x is new major release that is faster and even more powerful to the IDE 1.x.x. In addition to a more modern editor and a more responsive interface it includes advanced features to help users with their coding and debugging.

The following steps can guide you with using the offline IDE (you can choose either IDE 1.x.x or IDE 2.x):

1. Download and install the Arduino Software IDE:

Arduino IDE 1.x.x (Windows, Mac OS, Linux, Portable IDE for Windows and Linux, ChromeOS).

Arduino IDE 2.x

2. Connect your Arduino board to your device.

3. Open the Arduino Software (IDE).

The Arduino Integrated Development Environment - or Arduino Software (IDE) - connects to the Arduino boards to upload programs and communicate with them. Programs written using Arduino Software (IDE) are called sketches. These sketches are written in the text editor and are saved with the file extension. Ino.

Using the offline IDE 1.x.x

The editor contains the four main areas:

1. A **Toolbar with buttons** for common functions and a series of menus. The toolbar buttons allow you to verify and upload programs, create, open, and save sketches, and open the serial monitor.

2. The message area gives feedback while saving and exporting and displays errors.

3. The **text editor** for writing your code.

4. The **text console** displays text output by the Arduino Software (IDE), including complete error messages and other information.

The bottom right-hand corner of the window displays the configured board and serial port.

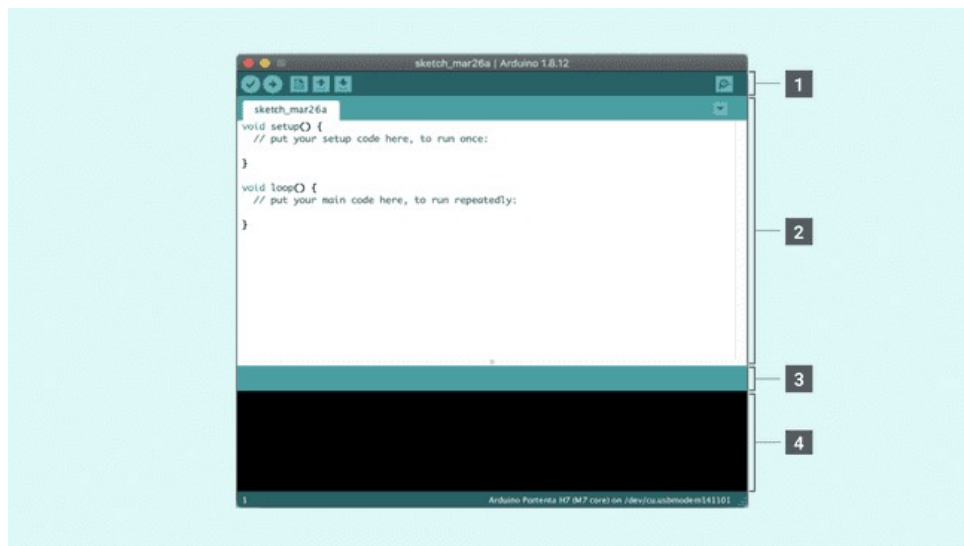


Figure 2: The Arduino Software IDE

[Using the Arduino Software \(IDE\) | Arduino Documentation](#)

Now that you are all set up, **let's try to make your board blink!**

5. **Connect your Arduino** or Genuine board to your computer.

6. Now, you need to **select the right core & board**. This is done by navigating to **Tools > Board > Arduino AVR Boards > Board**. Make sure you select the board that you are using. If you cannot find your board, you can add it from **Tools > Board > Boards Manager**.

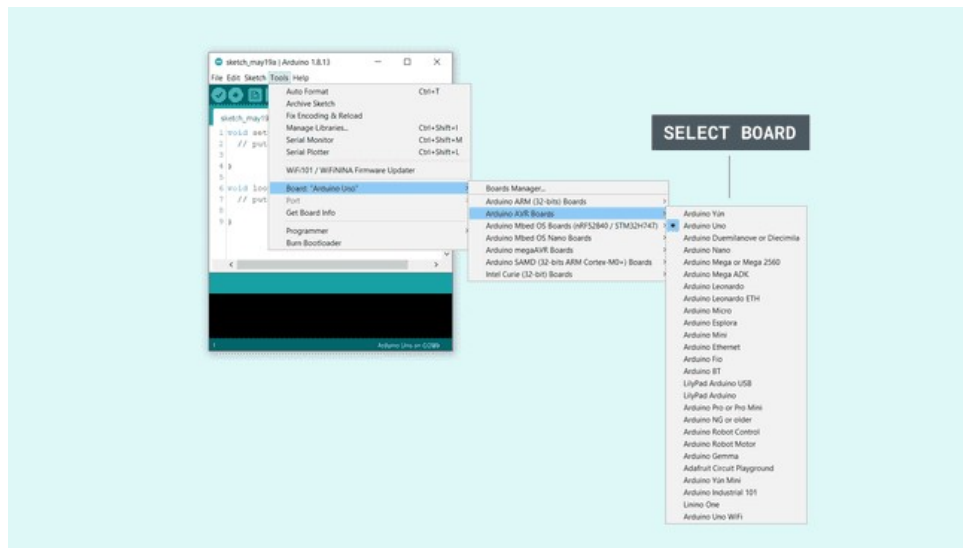


Figure 3: Selecting a board

### Using the Arduino Software (IDE) | Arduino Documentation

7. Now, let's make sure that your board is found by the computer, by **selecting the port**. This is simply done by navigating to **Tools > Port**, where you select your board from the list.



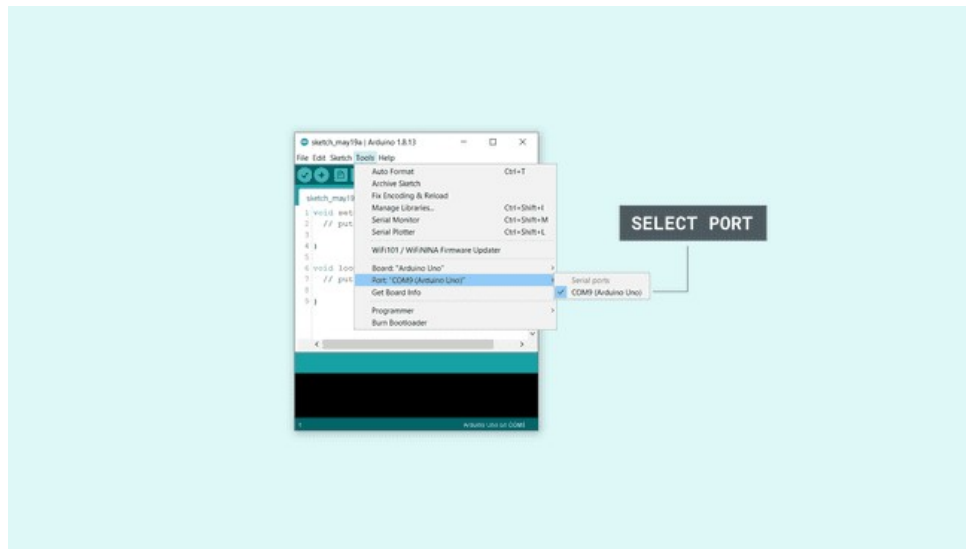


Figure 4: Selecting the port

[Using the Arduino Software \(IDE\) | Arduino Documentation](#)

**8. Let's try an example: navigate to File > Examples > 01. Basics > Blink.**

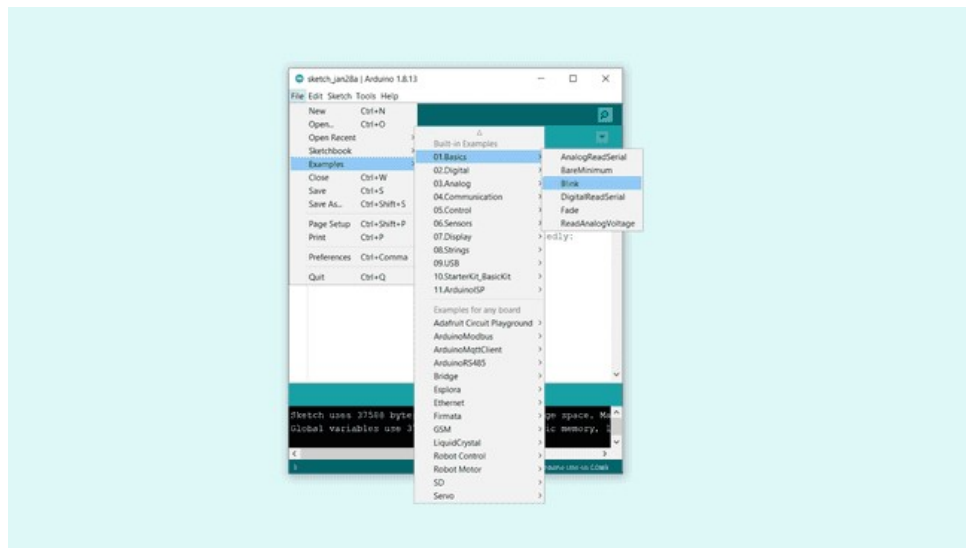


Figure 5: Opening an example

[Using the Arduino Software \(IDE\) | Arduino Documentation](#)

**9. To upload it to your board,** simply click on the arrow in the top left corner. This process takes a few seconds, and it is important to not disconnect the board

during this process. If the upload is successful, the message "Done uploading" will appear in the bottom output area.

**10.** Once the upload is complete, you should then see on your board the yellow LED with an L next to it start blinking. You can **adjust the speed of blinking** by changing the delay number in the parenthesis to 100 and upload the Blink sketch again. Now the LED should blink much faster.

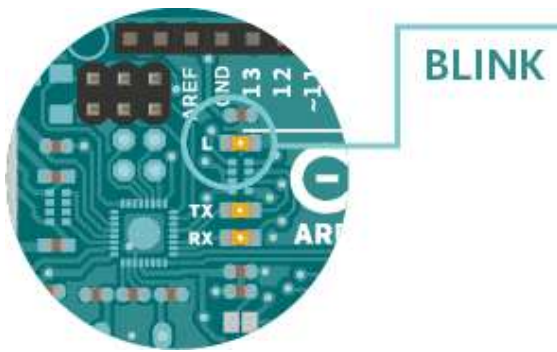


Figure 6: LED Blinking

## **Brief Overview of components**

### **1C2 LCD DISPLAY:**

A 1C2 LCD Display, often referred to as a 1602 LCD, is a popular module used in various electronic projects for displaying text. The "162" designation means the display has 16 columns and 2 rows, allowing for the display of 32 characters at a time. The "12C" likely refers to the 12C (Inter-Integrated Circuit) interface, which simplifies the wiring by reducing the number of pins needed to connect the LCD to a microcontroller, such as an Arduino or Raspberry Pi.

### **RFID MODULE RC522:**

The RFID Module RC522 is a popular and cost-effective module for integrating RFID (Radio Frequency Identification) technology into various projects. This module is widely used for creating RFID-based access control systems, contactless payment systems, inventory tracking, and more.

### **RFID tags**

Passive RFID tags are small, unpowered devices that transmit data wirelessly when exposed to an RFID reader's radio frequency (RF) signal. Unlike active RFID tags, which have their power source, passive tags rely on the energy supplied by the RFID reader to transmit information. They are cost-effective, lightweight, and commonly used in applications like inventory management, supply chain tracking, access control, and asset management.

### **SIM900 GSM module**

A GSM module is a specialized type of module used to enable communication between a microcontroller or a microprocessor and the GSM (Global System for Mobile Communications) network. These modules can send and receive SMS messages, make and receive calls, and connect to the internet using GPRS (General Packet Radio Service). They are widely used in applications like remote monitoring devices, home automation, vehicle tracking, and security systems.

## **Capabilities:**

SMS: Can send and receive SMS messages.

GPRS: Can connect to the internet using GPRS (though for this project, we'll focus on SMS).

Voice Calls: Capable of making and receiving voice calls (optional for this project).

## **Specifications:**

Operating Voltage: 4V to 7V

Communication: UART interface with AT commands

Frequency Bands: Dual-band 900/1800 MHz

Current Consumption: 1.5 mA (max)

Data Rate: Up to 112 kbps (theoretically)

## **Power adapter**

A power supply is an electrical device that supplies electric power to an electrical load. The main purpose of a power supply is to convert electric current from a source to the correct voltage, current, and frequency to power the load. As a result, power supplies are sometimes referred to as electric power converters.



Figure 7: I2C Display

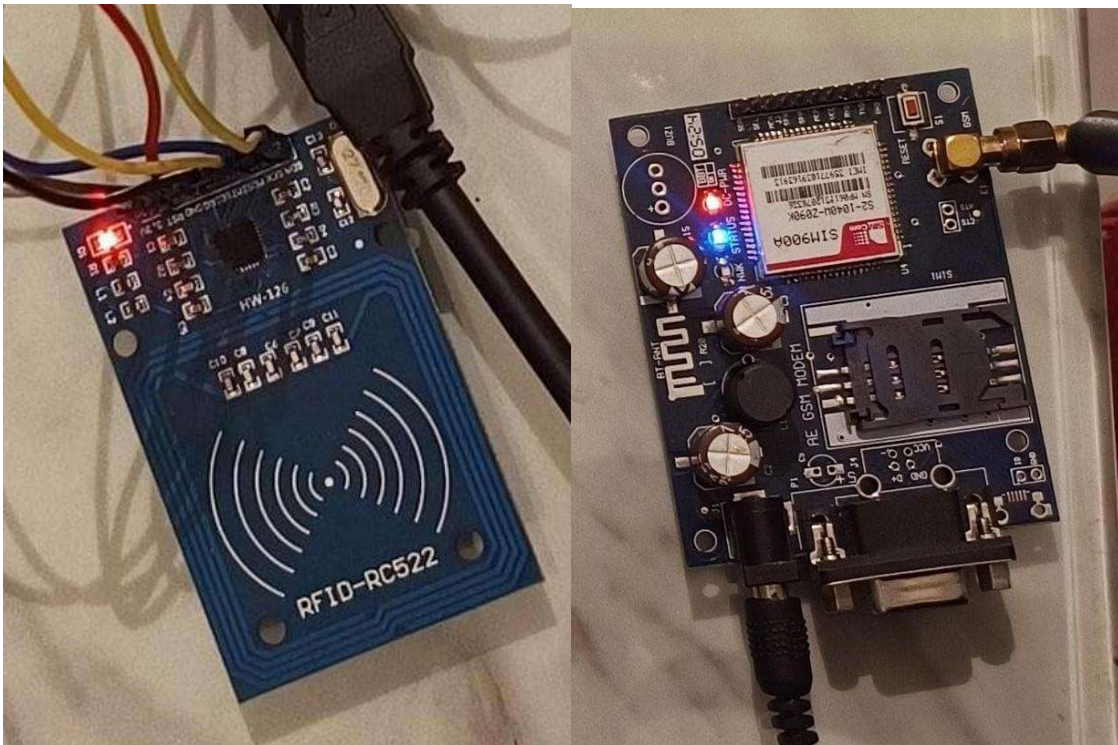


Figure 8: RFID module (left), Figure 9: Sim 900a (right).

## **SG90 Micro-servo motor**

The SG90 Micro Servo Motor is a small, lightweight servo motor commonly used in radio-controlled applications such as robots, helicopters, and aircraft. Compatible with Arduino.

## **4X4 Keypad**

General purpose keypad for data input. Compatible with Arduino.

## **Breadboard**

A breadboard (sometimes called a plug block) is used for building temporary circuits. It is useful to designers because it allows components to be removed and replaced easily. It is useful to the person who wants to build a circuit to demonstrate its action, then to reuse the components in another circuit.

## **Connecting wires/ Jumper wires**

A connecting wire allows travels the electric current from one point to another point without resistivity. Resistance of connecting wire should always be near zero. Copper wires have low resistance and are therefore suitable for low resistance.





Figure 10: 4X4 Keypad (left), Figure 11: Servo motor (right).

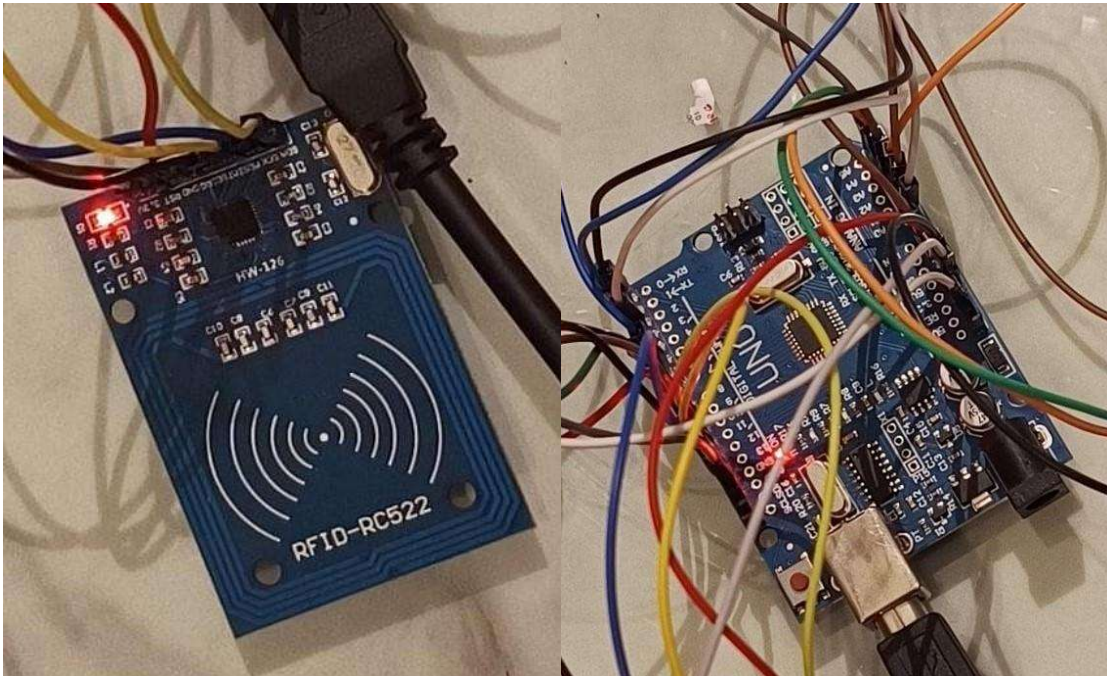


Figure 12: RFID module (left), Figure 13: Arduino (right).

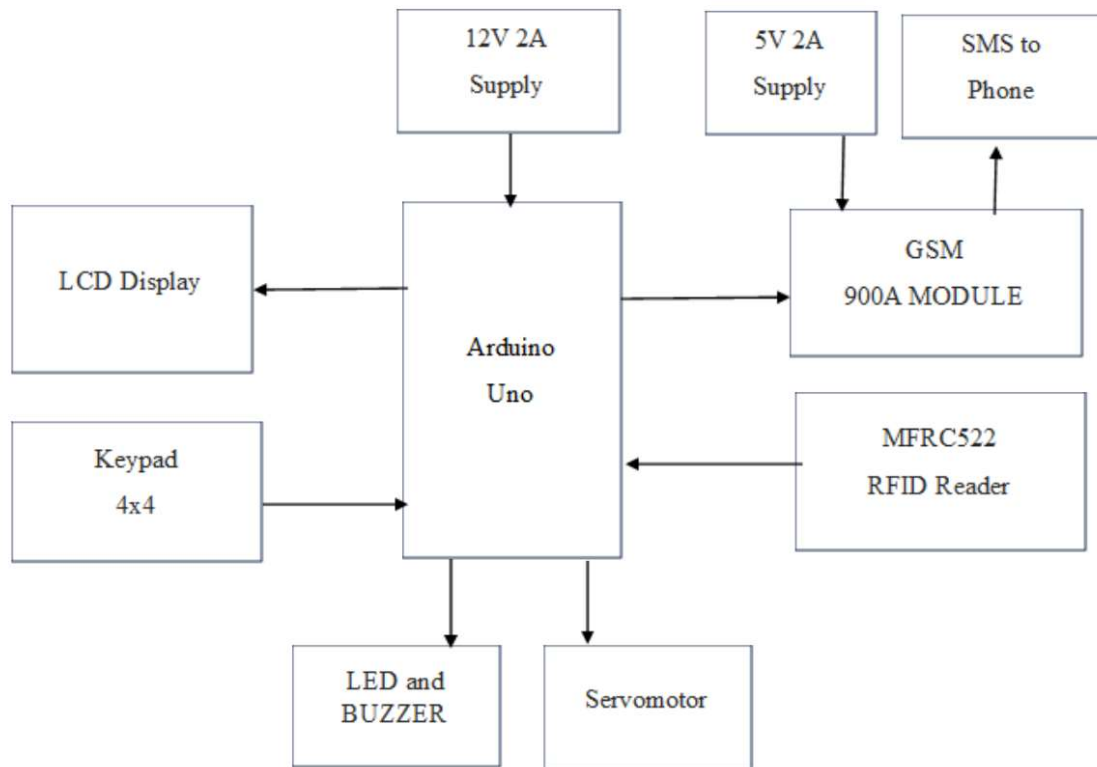


Chart 1: Block diagram



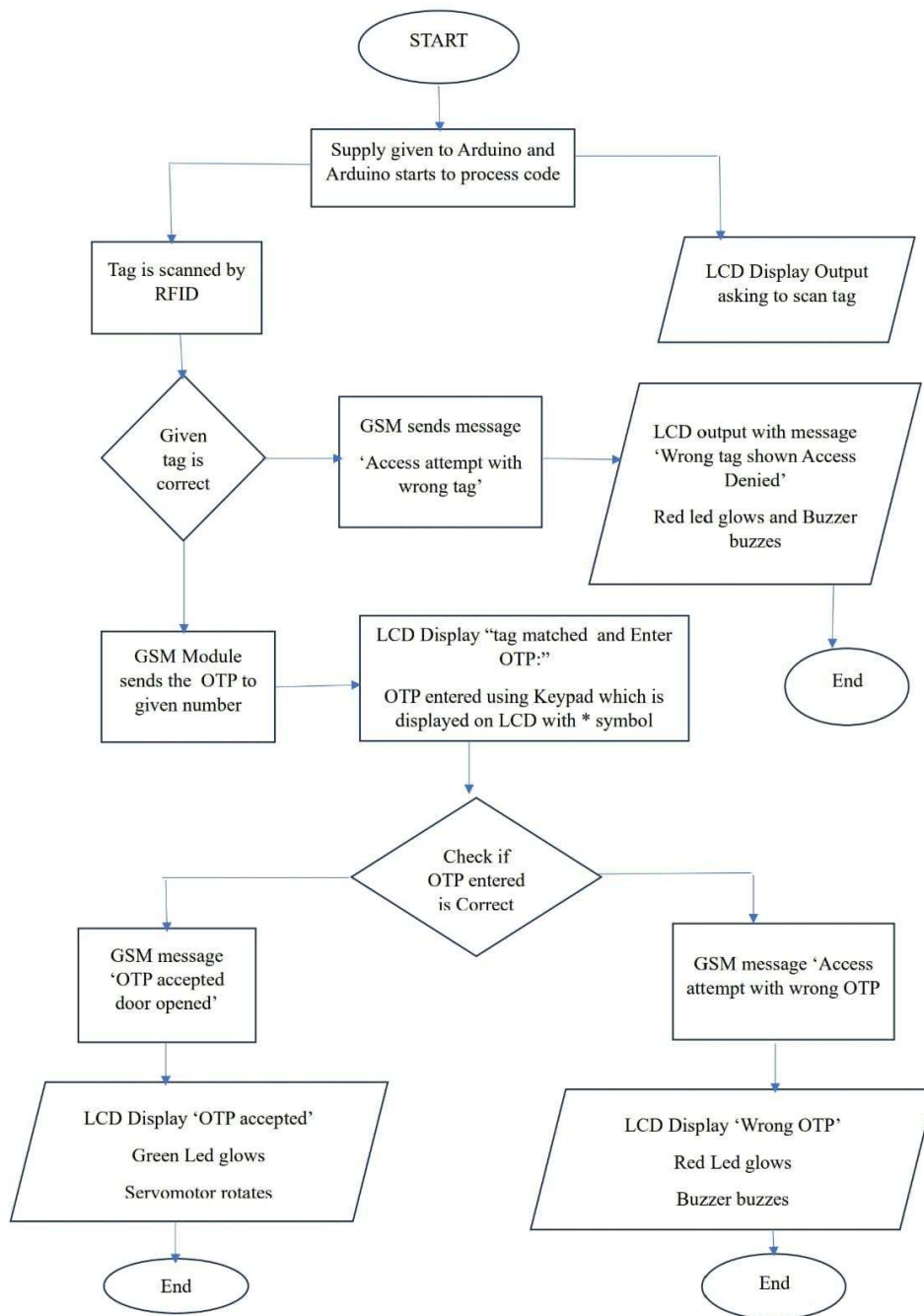


Chart 2: Flow Chart

# IMPLEMENTATION

## Circuit diagram/Hardware

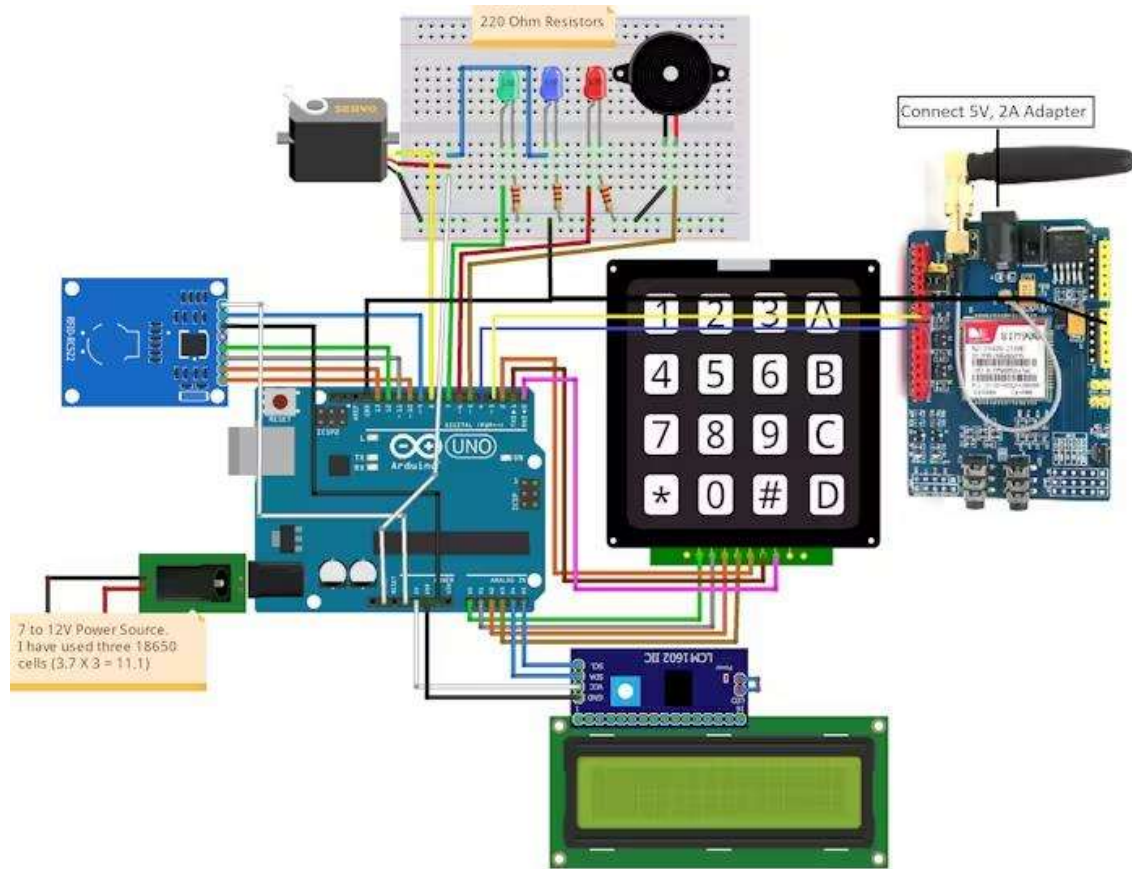


Figure 14: Circuit Diagram

Source: RFID and Keypad Door lock and Alert System Using Arduino - Hackster.io

## WORKING OF THE PROJECT

The system operates in 2 main stages: RFID Authentication, OTP Verification and alert system

### Stage 1: RFID Authentication

User Prompt: The LCD prompts the user to scan their RFID tag with the message "Scan Your Tag."

Tag Scanning:

- The RFID reader checks for a new card.
- If a card is detected, it reads the tag's UID.

UID Verification:

- The system compares the scanned UID with the stored authorized UID (tagUID).
- If the UID matches, an OTP is generated and sent to the user's phone via SMS.
- The LCD displays "Tag Matched" and prompts the user to "Enter OTP."

Access Denied:

- If the UID does not match, the LCD displays "Wrong Tag Shown" and "Access Denied."
- The buzzer and red LED are activated to signal the denial.
- An SMS alert is sent indicating an access attempt with the wrong tag.

### Stage 2: OTP Verification

Enter OTP:

- The user enters the received OTP using the keypad.
- Each keypress is displayed as an asterisk (\*) on the LCD for security.

OTP Validation:

- Once the user has entered the full OTP (4 digits), the system compares it with the generated OTP.

If the OTP matches:

- The LCD displays "OTP Accepted."
- The green LED is lit, and the servo motor unlocks the door.
- An SMS is sent confirming the OTP and door opening.
- The door remains open for a set duration (5 seconds), after which it locks again.
- The system resets and returns to RFID mode.

If the OTP does not match:

- The LCD displays "Wrong OTP."
- The buzzer and red LED are activated.
- An SMS alert is sent indicating an access attempt with the wrong OTP.
- The system resets and returns to RFID mode.

## Code/Software

Code used on arduino

```
// Include required libraries

#include <MFRC522.h>

#include <LiquidCrystal_I2C.h>

#include <Keypad.h>

#include <SoftwareSerial.h>

#include <Servo.h>

#include <SPI.h>

#include <stdlib.h>

#include <time.h>


// Create instances

SoftwareSerial SIM900(3, 4); // SoftwareSerial SIM900(Rx, Tx)

MFRC522 mfrc522(10, 9); // MFRC522 mfrc522(SS_PIN, RST_PIN)

LiquidCrystal_I2C lcd(0x27, 16, 2);

Servo sg90;


// Initialize Pins for LEDs, servo, and buzzer

constexpr uint8_t greenLed = 7;

constexpr uint8_t redLed = 6;

constexpr uint8_t servoPin = 8;

constexpr uint8_t buzzerPin = 5;
```

```

// Initialize OTP and RFID tag

String tagUID = "23 89 99 2F"; // String to store UID of tag. Change it with
your tag's UID

String otp; // Variable to store the OTP

char password[5]; // Variable to store user's OTP entry (5 for null terminator)

boolean RFIDMode = true; // Boolean to change modes

boolean NormalMode = true; // Boolean to change modes

char key_pressed = 0; // Variable to store incoming keys

uint8_t i = 0; // Variable used for counter


// Defining how many rows and columns our keypad have

const byte rows = 4;

const byte columns = 4;


// Keypad pin map

char hexaKeys[rows][columns] = {

    {'1', '2', '3', 'A'},

    {'4', '5', '6', 'B'},

    {'7', '8', '9', 'C'},

    {'*', '0', '#', 'D'}

};


// Initializing pins for keypad

byte row_pins[rows] = {A0, A1, A2, A3};

byte column_pins[columns] = {2, 1, 0};


// Create instance for keypad

```

```
Keypad keypad_key = Keypad(makeKeymap(hexaKeys), row_pins,  
column_pins, rows, columns);
```

```
void setup() {  
    // Seed random number generator  
    randomSeed(analogRead(0));  
  
    // Arduino Pin configuration  
    pinMode(buzzerPin, OUTPUT);  
    pinMode(redLed, OUTPUT);  
    pinMode(greenLed, OUTPUT);  
    sg90.attach(servoPin); // Declare pin 8 for servo  
    sg90.write(0); // Set initial position at 0 degrees  
  
    lcd.init(); // LCD screen  
    lcd.backlight();  
  
    SPI.begin(); // Init SPI bus  
    mfrc522.PCD_Init(); // Init MFRC522  
  
    // Initialize SIM900 GSM shield  
    SIM900.begin(19200);  
    SIM900.print("AT+CMGF=1\r");  
    delay(100);  
    SIM900.print("AT+CNMI=2,2,0,0,0\r");  
    delay(100);
```

```
    lcd.clear(); // Clear LCD screen  
    lcd.print("System Ready");  
}
```

```
void loop() {  
    if (!NormalMode) {  
        receive_message();  
    } else if (RFIDMode) {  
        handleRFIDMode();  
    } else {  
        handlePasswordMode();  
    }  
}
```

// Function to handle RFID mode

```
void handleRFIDMode() {  
    lcd.setCursor(0, 0);  
    lcd.print("  Door Lock");  
    lcd.setCursor(0, 1);  
    lcd.print(" Scan Your Tag ");  
  
    if (!mfrc522.PICC_IsNewCardPresent()) {  
        return;  
    }  
    if (!mfrc522.PICC_ReadCardSerial()) {  
        return;  
    }  
}
```



```

}

String tag = "";
for (byte j = 0; j < mfrc522.uid.size; j++) {
    tag.concat(String(mfrc522.uid.uidByte[j] < 0x10 ? " 0" : " "));
    tag.concat(String(mfrc522.uid.uidByte[j], HEX));
}
tag.toUpperCase();

if (tag.substring(1) == tagUID) {
    // Generate a random 4-digit OTP
    otp = String(random(1000, 9999));

    // Send OTP via SMS
    send_message("Your OTP is: " + otp);

    lcd.clear();
    lcd.print("Tag Matched");
    lcd.setCursor(0, 1);
    lcd.print("Enter OTP:");

    RFIDMode = false; // Disable RFID mode and switch to OTP mode
} else {
    lcd.clear();
    lcd.setCursor(0, 0);
    lcd.print("Wrong Tag Shown");
}

```

```

    lcd.setCursor(0, 1);
    lcd.print("Access Denied");
    digitalWrite(buzzerPin, HIGH);
    digitalWrite(redLed, HIGH);
    send_message("Access attempt with wrong tag.");
    delay(3000);
    digitalWrite(buzzerPin, LOW);
    digitalWrite(redLed, LOW);
    lcd.clear();
}
}

```

```

// Function to handle OTP entry mode
void handlePasswordMode() {
    key_pressed = keypad_key.getKey();
    if (key_pressed) {
        if (i < 4) {
            password[i++] = key_pressed;
            lcd.print("*");
        }
    }
    if (i == 4) {
        delay(200);

        password[i] = '\0'; // Null-terminate the string
    }
}

```

```

// Convert char array to String for comparison
String enteredOTP = String(password);

if (enteredOTP == otp) {
    lcd.clear();
    lcd.print("OTP Accepted");
    sg90.write(90); // Door Opened
    digitalWrite(greenLed, HIGH);
    send_message("OTP accepted. Door opened.");
    delay(5000); // Keep door open for 5 seconds
    sg90.write(0); // Door Closed
    lcd.clear();
    lcd.print("Access Granted");
    i = 0;
    RFIDMode = true; // Re-enable RFID mode
} else {
    lcd.clear();
    lcd.print("Wrong OTP");
    digitalWrite(buzzerPin, HIGH);
    digitalWrite(redLed, HIGH);
    send_message("Access attempt with wrong OTP.");
    delay(3000);
    digitalWrite(buzzerPin, LOW);
    digitalWrite(redLed, LOW);
    lcd.clear();
    i = 0;
}

```

```

    RFIDMode = true; // Re-enable RFID mode
}
}
}

// Receiving the message
void receive_message() {
    String incomingData;

    if (SIM900.available() > 0) {
        incomingData = SIM900.readString();
        delay(10);
    }

    if (incomingData.indexOf("open") >= 0) {
        sg90.write(90);
        NormalMode = true;
        send_message("Door opened by SMS command.");
        delay(10000); // Keep door open for 10 seconds
        sg90.write(0);
        lcd.clear();
        lcd.print("Door Opened");
        log_access("Door opened by SMS command.");
    } else if (incomingData.indexOf("close") >= 0) {
        NormalMode = false;
        send_message("System closed by SMS command.");
    }
}

```

```

    lcd.clear();
    lcd.print("System Closed");
}
}

// Function to send the message
void send_message(String message) {
    SIM900.println("AT+CMGF=1"); // Set the GSM Module in Text Mode
    delay(100);
    SIM900.println("AT+CMGS=\"" + 919490301470 "\""); // Replace with your
mobile number
    delay(100);
    SIM900.println(message); // The SMS text you want to send
    delay(100);
    SIM900.println((char)26); // ASCII code of CTRL+Z
    delay(100);

    if (SIM900.available()) {
        String response = SIM900.readString();
        Serial.println("GSM Response: " + response); // Print GSM response for
debugging
    }
}

// Function to log access events

```

```
void log_access(String event) {  
    lcd.clear();  
    lcd.setCursor(0, 0);  
    lcd.print("Event: ");  
    lcd.setCursor(0, 1);  
    lcd.print(event);  
    delay(3000); // Display log for 3 seconds  
    lcd.clear();  
    lcd.print("System Ready");  
}
```

# RESULT

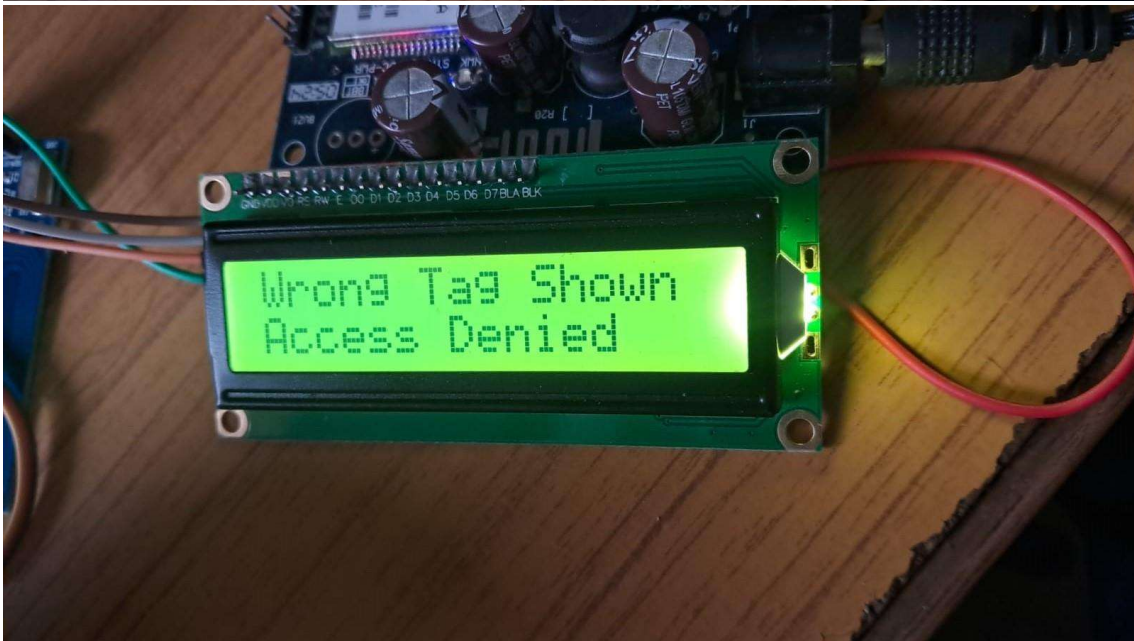


Figure 15:(top) photo is default screen when door is locked.

Figure 16:(bottom) photo is screen after the tag is wrong.

Access attempt with wrong tag.



Figure 17: (top) SMS alert sent to phone.

Figure 18: (middle) photo is of red LED glowing due wrong tag.

Figure 19: (bottom) photo is screen after the tag is matched.



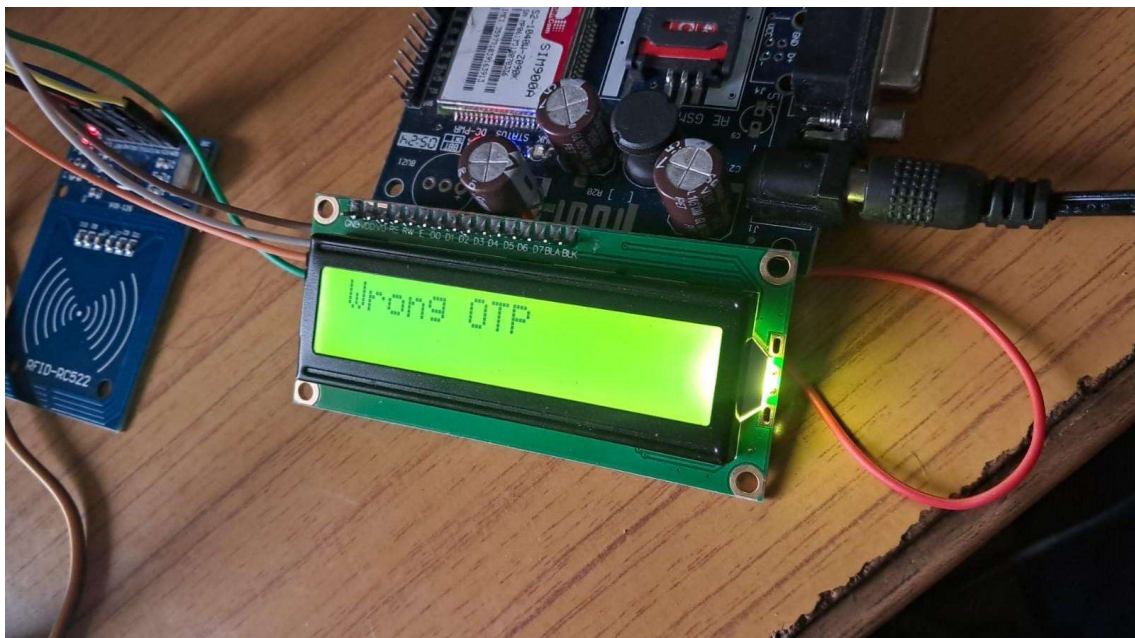
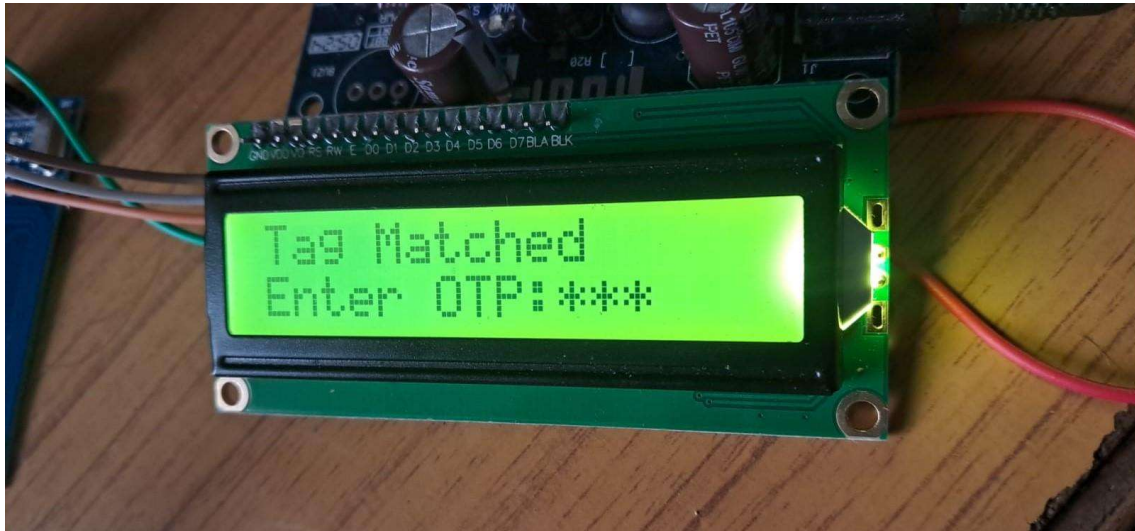


Figure 20:(top) photo is screen after the tag is matched and OTP being entered.

Figure 21: (bottom) photo is screen when the OTP entered is wrong.

Your OTP is: 9225

Access attempt with wrong OTP.

Your OTP is: 7250

Access attempt with wrong OTP.



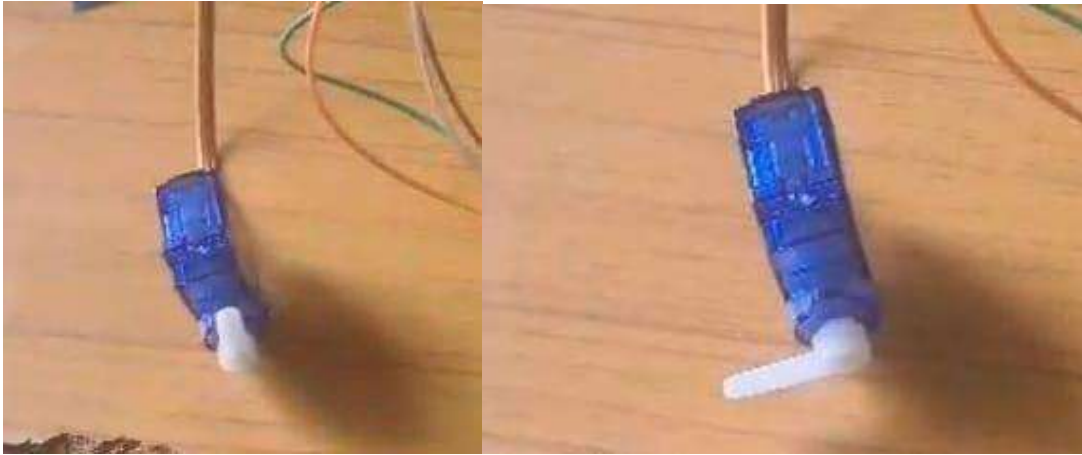
Figure 22: (top) photo is of alert being sent via SMS.

Figure 23: (bottom) photo is of red LED glowing due wrong OTP.



Figure 24: (top) photo of OTP accepted.

Figure 25: (bottom) photo is of green LED glowing due correct OTP.



Your OTP is: 6695

OTP accepted. Door opened.

Figure 26: (top left) Servo motor is dook locked position.

Figure 27: (top right) Servo motor is dook open position after correct OTP.

Figure 28: (bottom) SMS alert to know access has been granted.



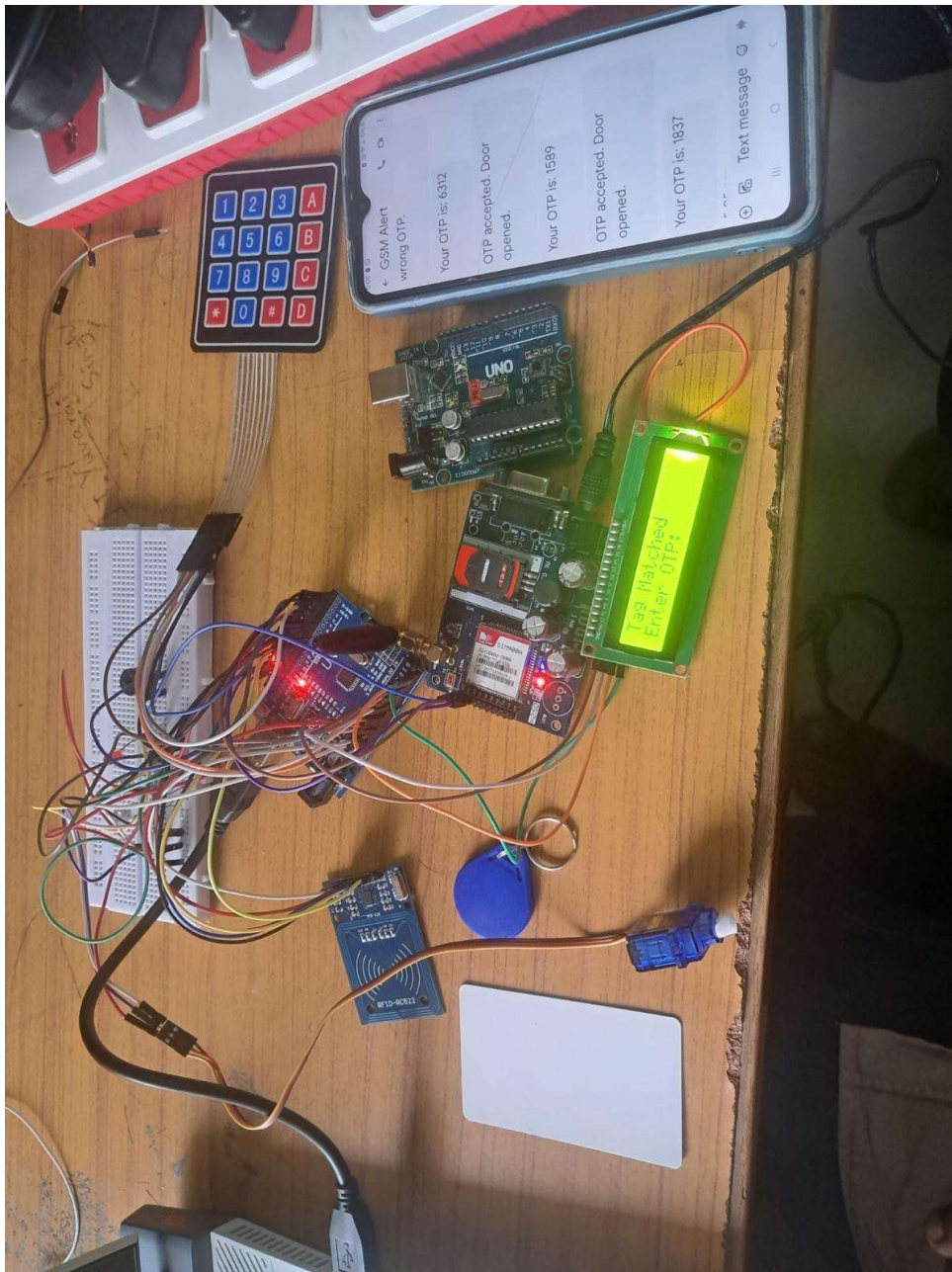


Figure 29: An image of all components assembled.

# CHALLENGES FACED

Integrating hardware and software for an RFID-based door lock system with OTP functionality involves several challenges in coding and hardware integration.

## Complexity in Software Design

- **Integration Logic:** Writing code to handle the interaction between RFID readers, OTP generators, and keypad inputs can be complex. Ensuring that the software handles all possible scenarios (e.g., successful access, denied access, system errors) is crucial.

## Error Handling and Debugging

- **Testing and Debugging:** Identifying and fixing bugs in a complex system with multiple components (hardware and software) can be time-consuming. Thorough testing is required to ensure reliability.

## User Interface Design

- **User Feedback:** Designing intuitive interfaces for users to interact with the system, including error messages, instructions for entering OTPs, and system status updates.
- **Responsiveness:** Ensuring that the software responds promptly to user actions, such as entering OTPs and RFID tags.

## Network and Communication

- **OTP Transmission:** Coding the logic for sending OTPs via SMS or other communication channels requires handling network issues, delays, and ensuring reliable delivery.

## Hardware Compatibility

- **Component Integration:** Ensuring that the RFID reader, keypad, and OTP generation hardware work seamlessly together. This might involve interfacing different hardware components and ensuring they communicate effectively.

- **Driver Issues:** Ensuring that the correct drivers are installed and configured for all hardware components.

### **Communication Protocols**

- **Interfacing:** Implementing the communication protocol between the hardware (e.g., RFID reader) and the software. This involves handling data exchange, signal processing, and ensuring accurate data transmission.
- **Timing Issues:** Managing timing and synchronization between hardware events (e.g., RFID detection) and software processes (e.g., OTP validation).

### **Hardware Calibration and Testing**

- **Calibration:** Calibrating sensors and readers to ensure accurate performance, such as reading RFID tags correctly.
- **Testing:** Rigorous testing of hardware components to ensure they perform reliably under various conditions.

# SCOPE FOR IMPROVEMENT

Enhancing an RFID-based door lock system integrated with OTP functionality can involve various improvements to increase security, usability, and reliability.

## Enhanced Security Features

- **Biometric Authentication:** Integrate biometric features such as fingerprint or facial recognition for added security, either as a primary method or in conjunction with OTP.
- **Multi-Factor Authentication (MFA):** Combine multiple authentication methods (e.g., RFID, OTP, and biometrics) for a higher level of security.
- **Encryption:** Use advanced encryption standards for OTP transmission and RFID communication to prevent interception and unauthorized access.

## User Experience Improvements

- **User-Friendly Interface:** Design an intuitive and responsive user interface for both the door lock system and any associated apps or web interfaces.
- **Backup Access Methods:** Provide alternative access methods (e.g., emergency override, backup codes) for situations where users may face issues with OTP delivery or RFID tags.
- **Customizable Settings:** Allow users to customize settings such as OTP validity duration, access schedules, and notification preferences.

## System Reliability and Performance

- **Redundancy and Failover:** Implement redundancy and failover mechanisms to ensure the system remains operational in case of hardware or network failures.
- **Real-Time Monitoring:** Develop real-time monitoring and alert systems to track system status, detect anomalies, and provide timely notifications to administrators.
- **Battery Management:** Ensure efficient power management for battery-operated components and provide alerts for low battery levels.



## **Integration and Scalability**

- **Cloud Integration:** Integrate with cloud services to manage and store access logs, user profiles, and OTP generation in a scalable manner.
- **API Support:** Provide APIs for integration with other security systems, building management systems, or IoT devices.
- **Scalability:** Design the system to scale easily for different sizes and types of installations, from small residential properties to large commercial buildings.

## **Advanced Features**

- **Geofencing:** Use geofencing to automate access control based on the location of authorized users' mobile devices.
- **AI and Machine Learning:** Implement AI and machine learning algorithms to detect unusual access patterns or potential security threats.
- **Voice Control:** Incorporate voice recognition or control via virtual assistants (e.g., Amazon Alexa, Google Assistant) for hands-free access management.

## **Improved Connectivity and Communication**

- **Multi-Network Support:** Support multiple communication networks (e.g., Wi-Fi, cellular, Bluetooth) to ensure reliable OTP delivery and system connectivity.
- **Offline Mode:** Provide functionality for offline access in case of network outages, using local storage for OTP verification and access control.

By focusing on these areas, you can enhance the RFID-based door lock system with OTP functionality, making it more secure, user-friendly, and adaptable to future needs and technological advancements.

# APPLICATIONS

## 1. Smart Homes and Home Automation:

- a. RFID door locks seamlessly integrate with smart home systems. Homeowners can control access remotely, receive notifications, and manage user permissions through mobile apps.
- b. Imagine arriving home with groceries—no need to fumble for keys! A simple swipe of an RFID card grants entry.
- c. Some advanced systems even allow temporary access for guests or service providers.

## 2. Energy Efficiency and Workplace Optimization:

- a. In office buildings, RFID locks contribute to energy conservation. Lights and HVAC systems can activate when an authorized user enters a room and deactivate when they leave.
- b. Conference rooms equipped with RFID locks can automatically adjust lighting and temperature based on occupancy.

## 3. Secure Document Storage and Confidential Areas:

- a. Legal firms, financial institutions, and government offices use RFID locks to secure sensitive documents.
- b. Access to confidential areas (such as server rooms or executive offices) is tightly controlled using personalized RFID cards.

## 4. Museum and Art Gallery Exhibits:

- a. RFID door locks protect valuable artifacts and artworks. Curators and authorized personnel can access display cases or storage rooms.
- b. The system logs each entry, enhancing security and accountability.

## 5. Locker Rooms and Fitness Centers:

- a. Gyms and sports facilities use RFID locks for locker rooms. Members swipe their cards for access.
- b. This eliminates the need for keys or combination locks, reducing the risk of lost keys or forgotten combinations.

## 6. Hospital and Laboratory Equipment Access:

- a. Hospitals secure medical equipment using RFID locks. Only authorized staff can access devices like ultrasound machines, defibrillators, or medication carts.
- b. Laboratories restrict entry to researchers with valid RFID credentials.

**7. Retail Stores and Inventory Control:**

- a. Retailers use RFID door locks to secure stockrooms and inventory areas.
- b. Employees can retrieve merchandise for restocking or customer orders by swiping their RFID cards.

**8. Theme Parks and Attractions:**

- a. RFID wristbands or cards grant access to rides, shows, and attractions.
- b. These systems enhance visitor experience by reducing wait times and enabling cashless transactions.

**9. Time and Attendance Tracking:**

- a. Some companies use RFID locks as part of their time and attendance systems.
- b. Employees swipe in upon arrival, and the system records their work hours.

**10. Automotive Industry:**

- a. Car dealerships secure vehicle keys using RFID locks. Salespeople access keys for test drives or customer demonstrations.
- b. Service centres use RFID systems to manage car keys during maintenance or repairs.

**11. Smart Lockers and Delivery Services:**

- a. E-commerce companies and package delivery services deploy RFID lockers.

12. Customers receive a unique code via SMS or app to open the locker and retrieve their packages.

**13. Event Management and Ticketing:**

- a. Concerts, conferences, and sports events use RFID wristbands or cards for entry.
- b. Attendees can quickly pass through gates, and event organizers gain insights into crowd movement.

## CONCLUSION

The "RFID based door lock system" using the RFID522 has been thoroughly designed, implemented and tested demonstrating it's effective and

Reliable system in door locks.

This system provides real time data and alerts which helps in safety

Key findings :

1)Real time alerts

The system which is used provides real time alerts which are very useful, there alerts are accurate and sent without delay

2)User friendly

The is easy to manage and not very difficult to understand and can be used by people of wide range of age

3) security

The system provides a secure door lock with negligible chance of breakthrough forcibly

4) stability

The is stable and works efficiently with optimum power consumption

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